

- 1 (a) Fig. 1.1 is a diagram of a human chromosome at a stage in mitosis.

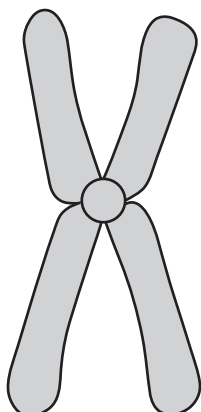


Fig. 1.1

- (i) The paragraph describes the structure of the chromosome shown in Fig. 1.1.

Complete the paragraph using the most appropriate term in each space provided.

The chromosome shown in Fig. 1.1 has two genetically identical joined at a The chromosome is composed of two DNA molecules, each wrapped around proteins known as proteins.

[3]

- (ii) State **one** stage during mitosis when the chromosome would appear as shown in Fig. 1.1.

..... [1]

- (iii) Suggest the role of ATP in the process of mitosis.

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..... [2]

(b) Prokaryotes divide by a process known as binary fission.

Fig. 1.2 shows some of the stages in binary fission.

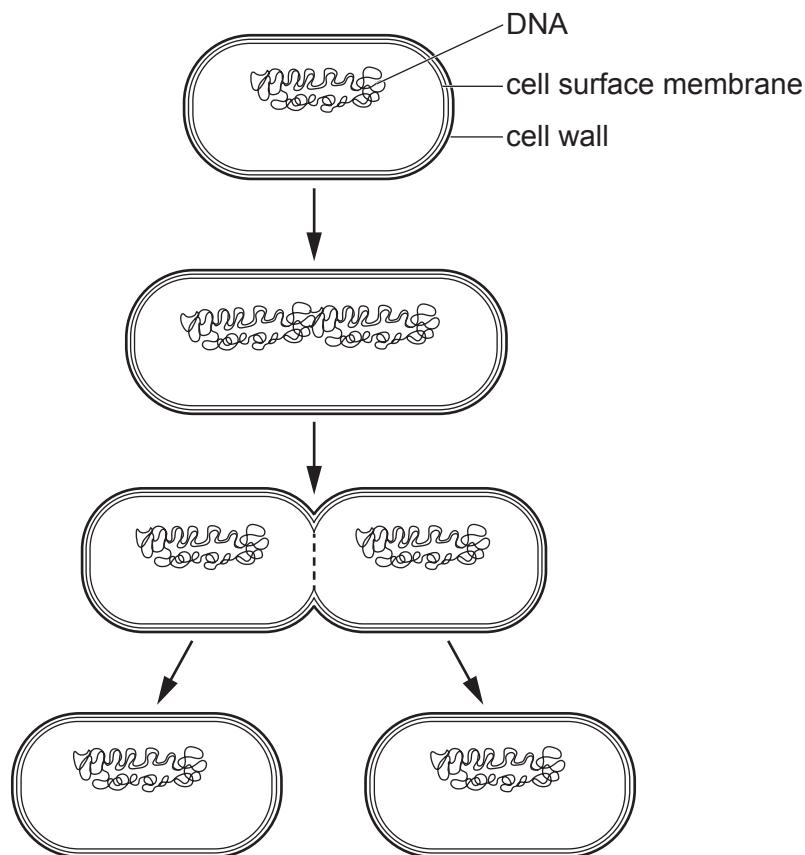


Fig. 1.2

With reference to Fig. 1.2, identify **two** events that occur during binary fission that do **not** occur during mitosis in human cells.

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..... [2]

[Total: 8]